SGA 6, Exposé V: Gamma operations and free lambda-rings

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Continuation of 2.9

We still suppose that K is binomial. Multiplication, for the law $\circ$, by $\lambda_t(a)$, where $a \in K$, defines a morphism of functors

$$\widehat{G}_K \longrightarrow \widehat{G}_K$$

which is characterized by its value on the element $1 + Xt$ of $G(K[X])$. Thus

$$\lambda_t(a) \circ (1 + Xt) = \lambda_{Xt}(a),$$

and hence

$$\lambda_t(a) \circ (1 + Xt) = (1 + Xt)^a = \exp(a \operatorname{Log}(1 + Xt)) = \sum_{n=0}^{\infty} \binom{a}{n} X^n t^n. \tag{2.9.3}$$

It follows that

$$\lambda_t(a) \circ \varphi = \varphi^a \quad (= \exp(a \operatorname{Log} \varphi)). \tag{2.9.4}$$

Indeed, the two members are additive functors which coincide on $1 + Xt$. From (2.9.3) and 1.15 one obtains that the universal polynomial of degree n defining multiplication by $\lambda_t(a)$ is isobaric of weight n. We denote it by $R_{n,a}$.

Finally, if A is a K-λ-algebra and $b \in A$, then

$$\lambda_t(ab) = (\lambda_t(b))^a. \tag{2.9.5}$$

This follows from $\lambda_t(ab) = \lambda_t(a) \circ \lambda_t(b)$ and from (2.9.4).

3. The operations γ

We still work over the category $\mathscr{C}_0$.

3.1. Let K be a λ-ring. We shall define from the operations λ^i new operations denoted γ^i. They are the operations which are adapted to Chern classes. If E is a vector bundle of rank d, the top Chern class of E is represented, in K-theoretic language, by the alternating sum of the exterior powers. To obtain the n-th class, one first replaces E by $E \oplus \mathbf{1}^{n-d}$, so that the same top-class recipe applies. This motivates the formula below.

Proposition 3.2. Let K be a λ-ring.

i) For every $x \in K$ and every $n \in \mathbb{N}$,

$$\lambda^n(x + n - 1) = (-1)^n \sum_{i=0}^{n} (-1)^i \lambda^i(x + n).$$

ii) If

$$\gamma^n(x) = \lambda^n(x + n - 1), \qquad \gamma_t(x) = \sum_{n \geq 0} \gamma^n(x) t^n,$$

then

$$\gamma_t(x) = \lambda_{t/(1-t)}(x), \qquad \text{hence} \qquad \lambda_t(x) = \gamma_{t/(1+t)}(x).$$

Thus knowledge of the operations γ is equivalent to knowledge of the operations λ.

Proof. For i),

$$\lambda^n(x+n-1) = \sum_{i=0}^{n} \lambda^i(x+n)\lambda^{n-i}(-1).$$

Since $\lambda_t(1) = 1+t$, one has $\lambda_t(-1) = 1/(1+t)$, and hence $\lambda^k(-1) = (-1)^k$, giving the desired relation. For ii),

$$\lambda_{t/(1-t)}(x) = \sum_{n\geq 0} \lambda^n(x)t^n(1-t)^{-n},$$

and

$$t^n(1-t)^{-n} = \sum_{p\geq n} \binom{p-1}{p-n} t^p.$$

The coefficient of t^p is therefore

$$\sum_{n=0}^{p} \lambda^n(x)\binom{p-1}{p-n} = \sum_{n=0}^{p} \lambda^n(x)\lambda^{p-n}(p-1) = \lambda^p(x+p-1).$$

3.3. If $f : K \to K'$ is a homomorphism of λ-rings, then f is a λ-homomorphism if and only if, for every $i \in \mathbb{N}$,

$$f(\gamma^i(x)) = \gamma^i(f(x)),$$

or equivalently

$$\gamma_t(f(x)) = \widehat{G}(f)(\gamma_t(x)).$$

3.4. From 3.2 one obtains

$$\gamma^0(x) = 1, \qquad \gamma^1(x) = x, \qquad \gamma^n(x+y) = \sum_{p+q=n} \gamma^p(x)\gamma^q(y). \tag{3.4.1}$$

Thus γ_t defines a new pre-λ-ring structure. It is not, in general, a λ-ring structure. Let ρ_t denote the automorphism of the functor $\widehat{G}$ induced by the change of variable

$$t \longmapsto \frac{t}{1-t}.$$

Then $\gamma_t = \rho_t\lambda_t$. Transporting the λ-ring functor structure of $\widehat{G}$ by ρ_t gives a second structure on $\widehat{G}$, for which $\gamma_t : K \to \widehat{G}(K)$ is a λ-homomorphism whenever K is a λ-ring.

3.5. The multiplication in this transported structure is the composite

$$\widehat{G} \times \widehat{G} \xrightarrow{\rho_t^{-1}\times\rho_t^{-1}} \widehat{G} \times \widehat{G} \xrightarrow{\circ} \widehat{G} \xrightarrow{\rho_t} \widehat{G}.$$

It will be denoted ∇. It is determined by

$$\begin{aligned}
(1+Xt)\nabla(1+Yt) &= \rho_t\Big(\rho_t^{-1}(1+Xt) \circ \rho_t^{-1}(1+Yt)\Big) \\
&= \rho_t\left(\frac{1+(X+1)t}{1+t} \circ \frac{1+(Y+1)t}{1+t}\right) \\
&= \rho_t\left(\frac{(1+((X+1)(Y+1))t)(1+t)}{(1+(X+1)t)(1+(Y+1)t)}\right) \\
&= \frac{1+(1+X+Y+XY)t}{(1+Xt)(1+Yt)}.
\end{aligned} \tag{3.5.1}$$

Thus, for $x, y \in K$,

$$\gamma_t(xy) = \gamma_t(x)\nabla\gamma_t(y). \tag{3.5.2}$$

The universal polynomial of degree n for this law will be denoted P'_n. Its isobaric components have weights between n and $2n$ in each of the two groups of variables, and

$$\gamma^n(xy) = P'_n(\gamma^1(x), \dots, \gamma^n(x), \gamma^1(y), \dots, \gamma^n(y)). \tag{3.5.3}$$

The unit for ∇ is $\gamma_t(1) = 1/(1-t)$.

3.6. The operations λ' of the transported structure are defined by

$$\widehat{G}$$

$$\begin{tikzcd} 1 + K[[t]]^+ \arrow[r, "\lambda_u"] \arrow[d, "\rho_t"'] & 1 + \{1 + K[[t]]^+\}[[u]]^+ \arrow[d, "\widehat{G}(\rho_t)"] \\ 1 + K[[t]]^+ \arrow[r, "\lambda'_u"'] & 1 + \{1 + K[[t]]^+\}[[u]]^+. \end{tikzcd} \tag{3.6.1}$$

If γ' denotes the γ-operations attached to these operations λ', then

$$\gamma'_u(1 + Xt) = \frac{1}{1-u} + \{1 + Xt\}u. \tag{3.6.2}$$

Let $Q'_{i,j}$ be the coefficient of t^j in $\gamma'^i(1 + X_1t + X_2t^2 + \cdots)$. It is a polynomial in $\mathbb{Z}[X_1, \dots, X_{ij}]$, whose isobaric components have weights from j to ij. For a λ-ring K and $x \in K$,

$$\gamma_t(\lambda^i(x)) = \lambda'^i(\gamma_t(x)), \qquad \gamma_t(\gamma^i(x)) = \gamma'^i(\gamma_t(x)), \tag{3.6.3}$$

and therefore

$$\gamma^j(\gamma^i(x)) = Q'_{i,j}(\gamma^1(x), \dots, \gamma^{ij}(x)). \tag{3.6.4}$$

The transported pre-λ-ring structure on $\widehat{G}$ is itself a λ-ring structure.

3.7. We denote by $\widehat{G}_\circ$ the functor $\widehat{G}$ with the structure of §2, and by $\widehat{G}_\nabla$ the same functor with the structure of §3. Giving a λ-ring structure on a ring A is equivalent either to giving a ring homomorphism $A \to \widehat{G}_\circ(A)$ satisfying the usual diagram for λ_t, or to giving a ring homomorphism $A \to \widehat{G}_\nabla(A)$ satisfying the corresponding diagram for γ_t. Explicitly, the two diagrams are

$$\begin{tikzcd} A \arrow[r, "\lambda_u"] \arrow[d, "\lambda_t"'] & \widehat{G}^u_\circ(A) \arrow[d, "\widehat{G}^u_\circ(\lambda_t)"] \\ \widehat{G}^t_\circ(A) \arrow[r, "\lambda_u"'] & \widehat{G}^u_\circ(\widehat{G}^t_\circ(A)) \end{tikzcd} \tag{3.7.1}$$

and

$$\begin{tikzcd} A \arrow[r, "\gamma_u"] \arrow[d, "\gamma_t"'] & \widehat{G}^u_\nabla(A) \arrow[d, "\widehat{G}^u_\nabla(\gamma_t)"] \\ \widehat{G}^t_\nabla(A) \arrow[r, "\gamma'_u"'] & \widehat{G}^u_\nabla(\widehat{G}^t_\nabla(A)). \end{tikzcd} \tag{3.7.2}$$

3.8. Let K be a binomial ring. From (2.6.1), for $x \in K$,

$$\gamma_t(x) = \left(\frac{1}{1-t}\right)^x = \exp(-x \operatorname{Log}(1-t)). \tag{3.8.1}$$

For $a \in K$, multiplication by $\gamma_t(a)$ for the law ∇ is characterized by its value on $1 + Xt$:

$$\gamma_t(a)\nabla(1 + Xt) = (1 + Xt)^a = \exp(a \operatorname{Log}(1 + Xt)) = \sum_{n=0}^{\infty} \binom{a}{n} X^n t^n. \tag{3.8.2}$$

Thus the universal polynomial of degree n defining this multiplication is again $R_{n,a}$.

Definition 3.9. Let K be a λ-ring. An *augmented K-λ-algebra* is a K-λ-algebra A, together with a λ-homomorphism of K-algebras

$$\epsilon : A \longrightarrow K.$$

Morphisms are λ-homomorphisms compatible with the augmentations.

Example 3.9.1. Let (X, A) be a locally ringed topos. The Grothendieck group $K^\circ(X)$ of locally free finite-type A-modules has a natural pre-λ-ring structure by exterior powers. There is a natural λ-homomorphism from $H^0(X, \mathbb{Z})$ to $K^\circ(X)$: a section a of the constant sheaf $\mathbb{Z}$ partitions the final object into opens U_n on which $a = n$, and is sent to the class of the locally free module which is A^n on U_n for $n \geq 0$, minus the analogous class on the negative pieces. The rank map is the augmentation. This is the augmented structure used below.

3.10. Let A be an augmented K-λ-algebra. For $n > 0$, define $\mathrm{Fil}^n A$ to be the ideal generated by products

$$\gamma^{i_1}(x_1) \cdots \gamma^{i_k}(x_k), \qquad i_1 + \cdots + i_k \geq n, \qquad \epsilon(x_i) = 0. \tag{3.10.1}$$

If A is generated as an abelian group by (x_α), then $\mathrm{Fil}^n A$ is generated over K by

$$\gamma^{i_1}(x_{\alpha_1} - \epsilon(x_{\alpha_1})) \cdots \gamma^{i_k}(x_{\alpha_k} - \epsilon(x_{\alpha_k})), \qquad i_1 + \cdots + i_k \geq n. \tag{3.10.2}$$

The filtration is decreasing and multiplicative, and

$$\mathrm{Fil}^0 A = A, \qquad \mathrm{Fil}^1 A = \mathrm{Ker}\,\epsilon.$$

It is functorial for homomorphisms of augmented K-λ-algebras.

Proposition 3.11. Let A be a λ-ring and let $y_1, \ldots, y_n$ be elements of A such that $\gamma^j(y_i) = 0$ for $j \geq 2$. Put $x = \sum_i y_i$. Then $\gamma^k(x - n) = 0$ for $k > n$, and, for $1 \leq k \leq n$, $\gamma^k(x - n)$ is the k-th elementary symmetric function of the $y_i - 1$. This follows from

$$\gamma_t(x - n) = \prod_{i=1}^{n} \gamma_t(y_i - 1).$$

4. λ-rings generated by generators and relations

4.1. Let K be a λ-ring and let $(X_i)_{i \in I}$ be indeterminates. Put

$$A = K[X_i, {}^n X_i]_{i \in I,\ n \geq 2}.$$

Define

$$\lambda_t(X_i) = 1 + X_i t + \sum_{n \geq 2} {}^n X_i t^n, \qquad \lambda_t({}^n X_i) = \lambda^n(\lambda_t(X_i)). \tag{4.1.1}$$

This gives a λ-ring structure on A extending that of K.

Proposition 4.2. For every K-λ-algebra B and every family $(b_i)_{i \in I}$ in B, there exists a unique K-λ-homomorphism $A \to B$ carrying X_i to b_i. Necessarily it carries ${}^n X_i$ to $\lambda^n(b_i)$, and this condition is sufficient.

4.3. The same construction may be made using γ's. Let

$$A' = K[X_i, {}^\gamma_n X_i]_{i \in I,\ n \geq 2},$$

and define

$$\gamma_t(X_i) = 1 + X_i t + \sum_{n\geq 2} {}^{\gamma}_{n}X_i t^n, \qquad \gamma_t({}^{\gamma}_{n}X_i) = \gamma'^n(\gamma_t(X_i)). \tag{4.3.1}$$

This gives a λ-ring satisfying the same universal property, hence canonically isomorphic to A.

Definition 4.4. The preceding λ-ring is the *free λ-ring* over K generated by the family (X_i).

Definition 4.5. An ideal J of a λ-ring A is a λ-*ideal* if $\lambda^n(x) \in J$ for every $x \in J$ and $n > 1$. Kernels of λ-homomorphisms are λ-ideals; conversely, if J is a λ-ideal, A/J receives a unique λ-ring structure. The intersection of λ-ideals is a λ-ideal. Thus every family of elements has a smallest containing λ-ideal, generated by the elements and all their iterated λ^n's.

Definition 4.6. The quotient of the free λ-ring by the λ-ideal generated by prescribed relations (P_i) is called the λ-ring generated by the generators (X_i) subject to the relations (P_i). Equivalently, one may formulate the same construction in terms of γ-operations.

Proposition 4.7. This quotient has the expected universal property: every family (b_i) in a K-λ-algebra satisfying the relations (P_i) determines a unique K-λ-homomorphism from the quotient to that algebra.

Proposition 4.8. Let $n \geq 0$. In the free λ-ring over K on one generator X, the ideal generated by the $\lambda^k X$, $k > n$, is a λ-ideal. Indeed, after quotienting, one has only to check that if $\varphi \in \widehat{G}(C)$ has no terms of degree $> n$, then $\lambda^k(\varphi) = 0$ for $k > n$. By the splitting principle, it is enough to write

$$\varphi = \prod_{i=1}^{n}(1 + T_i t),$$

and then

$$\lambda_u(\varphi) = \prod_{i=1}^{n}(1 + \{1 + T_i t\}u),$$

which has degree n in u.

Corollary 4.9. The λ-ring generated over K by one element X, subject to $\lambda^k(X) = 0$ for $k > n$, is

$$K[X, \lambda^2 X, \ldots, \lambda^n X].$$

More generally, for a family (X_i) with bounds (n_i), the quotient subject to $\lambda^k(X_i) = 0$ for $k > n_i$ is the polynomial ring generated by $X_i, \lambda^2 X_i$,
$ldots, \lambda^{n_i} X_i$.

4.10. Let K be binomial and let A be an augmented K-λ-algebra. Put

$$B = A[X, \gamma^2 X, \gamma^3 X, \ldots],$$

the free A-λ-algebra on X, written in γ-coordinates. Extend the augmentation by $\epsilon(X) = 0$, hence $\epsilon(\gamma^i X) = 0$ for $i > 0$. Let M_k be the set of monomials in the variables $\gamma^j X$, counted with weight j, of total weight k, and set $M_0 = \{1\}$.

Lemma 4.11. If $x \in \mathrm{Fil}^p A$, then every coefficient of $\gamma^n(x + X)$ which is homogeneous of weight k in the $\gamma^j X$ belongs to $\mathrm{Fil}^{p+n-k} A$.

Proposition 4.12. For every $k \geq 0$,

$$\mathrm{Fil}^k B = \mathrm{Fil}^k A + \mathrm{Fil}^{k-1} A\, M_1 + \cdots + \mathrm{Fil}^1 A\, M_{k-1} + \mathrm{Fil}^0 A\, M_k + B\, M_{k+1}. \tag{4.12.1}$$

The inclusion from right to left follows from Lemma 4.11. The reverse inclusion is obtained by writing the generators of $\mathrm{Fil}^k B$ as products of γ-operations and reducing, term by term, to the displayed weights.

Remarks 4.13. The proof gives the same assertion for a free λ-algebra on any finite or infinite family of generators, with the evident multi-weight. It is also unchanged when one uses the λ-coordinates, since λ and γ coordinates differ by triangular universal formulas.

Corollary 4.14. Passing to the associated graded ring, one has

$$\operatorname{Gr}^k B \simeq \operatorname{Gr}^k A \oplus \operatorname{Gr}^{k-1} A\, M_1 \oplus \cdots \oplus \operatorname{Gr}^0 A\, M_k. \tag{4.14.1}$$

Corollary 4.15. If $B' = K[X, \gamma^2 X, \ldots, \gamma^n X]$ is the K-λ-algebra generated by one element satisfying $\gamma^k(X) = 0$ for $k > n$, then the λ-ring filtration of B' is exactly the weight filtration in the variables $\gamma^j X$.

5. The elements $\lambda^p(N, x)$

This section will be used only in Exposé VII, §4, in the study of regular immersions; it may be omitted on first reading.

5.1. Let $d \geq 1$. Let A be the $\mathbb{Z}$-λ-algebra generated by N and x, subject to the relations $\lambda^i(N) = 0$ for $i > d$. Thus

$$A = \mathbb{Z}[N, \ldots, \lambda^d N; x, \ldots, \lambda^j x, \ldots].$$

Put

$$\lambda_{-1}(N) = \sum_{i=0}^{d} (-1)^i \lambda^i(N). \tag{5.1.1}$$

Lemma 5.2. For every $p \geq 1$, the element $\lambda^p(x\lambda_{-1}(N))$ is divisible by $\lambda_{-1}(N)$.

It is enough, by isobaricity, to prove this for $\lambda^p(\lambda_{-1}(N))$. Let B be the λ-ring generated by $N_1, \ldots, N_d$, with $\lambda^i(N_j) = 0$ for $i \geq 2$. The map $N \mapsto \sum_i N_i$ identifies A with the symmetric polynomials in the N_i, and

$$\lambda_{-1}(N) = \prod_{i=1}^{d} (1 - N_i).$$

If some $N_i = 1$, then $\lambda_t(\lambda_{-1}(N)) = \lambda_t(0) = 1$, so every positive coefficient vanishes. Thus each $\lambda^p(\lambda_{-1}(N))$ is divisible by every $1 - N_i$, hence by their product.

Definition 5.3. Since A is integral, define $\lambda^p(N, x)$ by

$$\lambda^p(N, x)\lambda_{-1}(N) = \lambda^p(x\lambda_{-1}(N)), \qquad p \geq 1. \tag{5.3.1}$$

Thus $\lambda^p(N, x)$ is a polynomial in the $\lambda^i(x)$ and $\lambda^j(N)$, isobaric of weight p in the $\lambda^i(x)$. If a λ-ring B contains an element N' with $\lambda^k(N') = 0$ for $k > d$, then $\lambda^p(N', x')$ is defined by substitution; it is independent of the chosen bound d.

5.4. In the λ-ring generated by N, x, y, subject to $\lambda^k(N) = 0$ for $k > d$, the formula (5.3.1) gives

$$\lambda^p(N, x + y) = \lambda^p(N, x) + \lambda^p(N, y) + \sum_{\substack{i+j=p \\ i,j>0}} \lambda^i(N, x)\lambda^j(N, y)\lambda_{-1}(N). \tag{5.4.1}$$

If N, N', x satisfy the corresponding finite-rank hypotheses, then

$$\lambda^p(N + N', x)\lambda_{-1}(N) = \lambda^p(N', x\lambda_{-1}(N)). \tag{5.4.2}$$

5.5. Let A be a λ-ring and let $N \in A$ satisfy $\lambda^k(N) = 0$ for $k > d$. Define on the underlying abelian group of A a new multiplication by

$$x \cdot_N y = xy\lambda_{-1}(N).$$

After adjoining a unit, one obtains a ring $\mathbb{Z} \times A$ with

$$(n, x)(m, y) = (nm, ny + mx + x \cdot_N y).$$

The formulas

$$\lambda_t(n, x) = (1 + t)^n \left(1 + \sum_{p \geq 1} \lambda^p(N, x)t^p\right) \tag{5.5.1}$$

define a pre-λ-ring structure. In fact, when A is a λ-ring, this is a λ-ring structure; it is the one transported from the original structure by multiplication by $\lambda_{-1}(N)$. We denote this augmented λ-ring by $\widehat{A}_N$. Its γ-operations satisfy

$$\gamma^p(N, x)\lambda_{-1}(N) = \gamma^p(x\lambda_{-1}(N)). \tag{5.5.2}$$

5.6. Suppose that, in addition, L is an element such that $\lambda^k(L) = 0$ for $k > 1$. Then $\lambda^p(N+L, x)$ is the coefficient of t^p in the series

$$\frac{\sum_{k \geq 1} \lambda^k(N, x)(1 + L + \cdots + L^{k-1})t^k}{1 + \sum_{k \geq 1} \lambda^k(N, x)\lambda_{-1}(N)L^k t^k}. \tag{5.6.1}$$

This is obtained by applying (5.4.2) to $N + L$ and using $\lambda_{-1}(L) = 1 - L$.

Proposition 5.7. Let K be a binomial ring, A an augmented K-λ-algebra, and $N \in A$ such that $\lambda^k(N) = 0$ for $k > d$. Then, for every $x \in A$ and every $p \geq 1$,

$$\gamma^p(N, x) \in \mathrm{Fil}^{p-d} A. \tag{5.7.1}$$

Indeed, let A_0 be the K-λ-algebra generated by N_0', X_0', with $\lambda^k(N_0') = 0$ for $k > d$, and augmented by $\epsilon(N_0') = \epsilon(X_0') = 0$. This is a polynomial algebra over K in the variables $\gamma^i(N_0')$, $1 \leq i \leq d$, and $\gamma^j(X_0')$, $j \geq 1$, and its λ-filtration is the weight filtration. Put

$$N_0 = N_0' + d, \qquad X_0 = X_0' + \epsilon(x).$$

For N_0, X_0, the assertion is equivalent to

$$\gamma^p(N_0, X_0)\gamma^d(N_0') \in \mathrm{Fil}^p A_0.$$

Since $\gamma^d(N_0') = (-1)^d\lambda_{-1}(N_0)$, this becomes

$$(-1)^d\gamma^p(X_0\gamma^d(N_0')) \in \mathrm{Fil}^p A_0,$$

which is clear because $\gamma^d(N_0')$ has zero augmentation. A K-λ-homomorphism $A_0 \to A$, compatible with augmentations and sending N_0' to $N - d$ and X_0' to $x - \epsilon(x)$, gives the stated result.

6. Chern ring: opening

Throughout this section the base ring is a fixed binomial ring K. We use the functor G of 1.1, and return to the notation of §1: G denotes the functor there defined and G_i its restriction to $\mathscr{C}_0$.

6.1. In 3.5 and 3.6 we defined morphisms of functors

$$G_i \times G_i \xrightarrow{\star} G_i, \qquad G_i^\circ \xrightarrow{\gamma'} G_i^\circ \circ G_S, \qquad G_i \xrightarrow{a} G_i,$$

where the last map is multiplication, for the law ∇, by $\gamma_t(a)$ with $a \in K$. By 1.9 these give morphisms

$$G \times G \xrightarrow{\star} G, \qquad G \xrightarrow{\gamma'} G \circ G_S, \qquad G \xrightarrow{a} G.$$

The isobaricity of the universal polynomials involved implies that the first and third images lie in G, while the second lies in the subgroup of formal series in u whose non-zero terms have the prescribed degrees.